Federico García

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RESEARCH EXPERIENCE

CONICET Researcher, IAR — *Argentina*
Since Dec 2025 (Independent Researcher) | Sep 2021 - Dec 2025 (Assistant Researcher)

Theoretical and observational studies of compact and extended remnants in different astrophysical scenarios.

Guest Researcher, Yunnan Observatories — *China*
May - June 2026 (Host: Prof. G. Zhang)

X-ray binaries, quasi-periodic oscillations and spectral-timing techniques

Guest Researcher, University of Groningen — *the Netherlands*
May - August 2023 (Host: Prof. M. Méndez)

X-ray binaries, spectral-timing models and observations

Guest Researcher, Université Paris Cité — *France*
July - August 2022 (Host: Prof. S. Chaty)

X-ray binaries, formation channels, Gravitational-wave progenitors

Postdoc, University of Groningen — *the Netherlands*
September 2019 - August 2021 (Supervisor: Prof. M. Méndez)

Spectral-timing Comptonization models and observations of X-ray binary systems

Postdoc LABEX, University of Paris — *France*
September 2017 - August 2019 (Supervisor: Prof. S. Chaty)

Gravitational-wave progenitors: from massive stellar-binary systems to BH binaries

Postdoc CONICET, IAR — *Argentina*
April 2016 - August 2017 (Supervisor: Prof. J.A. Combi)

Investigations of neutron stars and supernova remnants in the X-ray band

Ph.D. Fellow of CONICET, IAR — *Argentina*
April 2011 - March 2016 (Supervisor: Prof. J.A. Combi)

Theoretical and observational studies of neutron stars in several astrophysical scenarios

TEACHING EXPERIENCE

Head Teaching Assistant (JTP), UNLP — *Argentina*
Since September 2021 | August 2015 - August 2017

Subject: Introduction to X-ray Astronomy

Teaching Assistant (graduated), UNLP — *Argentina*
September 2011 - August 2017

Subjects: Physics 1, 2, 3, Elements of Theoretical Astrophysics.

Teaching Assistant (student), UNLP — *Argentina*
February 2007 - July 2011

Subjects: Calculus I, Algebra, Astronomy, Computation, Introductory Math FCAGLP.

FORMATION

PhD. in Astronomy UNLP (Arg. | 2011-2016).
Thesis: Theoretical and observational studies of neutron stars in different astrophysical scenarios.
Supervisor: Prof. J.A. Combi.

M. Sc. in Astronomy UNLP (Arg. | 2004-2011).
Thesis: Magnetic-field evolution in accreting neutron stars.
Supervisor: Dr. D. Aguilera.

LANGUAGES

Spanish (native)
English (fluid)
French (intermediate)

AWARDS

2024 - Scientific Labour Young Prize UNLP, Argentina.
2023 - Enrique Gaviola Prize, National Academy of Sciences.
2022 - Young Prize, National Academy of Physical Sciences.
2016 - Distinguished Ph.D. Student UNLP, Argentina
2012 - Distinguished M. Sc. Student UNLP, Argentina

PUBLICATIONS

Papers published in refereed international journals: 99
(Nature 1, Nat. Astronomy 1, A&A 46, MNRAS 36, ApJ 7, GRG 2, PRD 2, PRX 2, JPhG 1, JHEA 1)
Papers under review: 2 **Astronomer Telegrams:** 9 (ATel)
Papers published in refereed national journals: 15 (BAAA), 3 (RvMxA)
Proceedings published: 4 (IAUS), 4 (EAS, AAS, APS, HEAD)

Papers as First Author (11), **Second Author** (23), **Third Author** (23).

Full Publication List in ADS (8308 citations, H-index: 29, i10-index: 62)
Google Scholar Profile (9153 citations, H-index: 31, i10-index: 67)

SEMINARS AND COLLOQUIA

Colloquia and Seminars out of Argentina: 32 **Colloquia and Seminars in Argentina:** 18

ATTENDANCE TO SCIENTIFIC MEETINGS AND PHD SCHOOLS

International Scientific Meetings: 23 **National Scientific Meetings:** 18
International Ph.D. Schools: 8

INTERNATIONAL RESEARCH VISITS

France *1*, **Germany** *1*, **Argentina** *1*, **Spain** *6*, **Italy** *1*, **Japan** *1*, **Netherlands** *4*, **UK** *2*, **Czech R.** *1*, **UAE** *1*, **China** *1*

SUPERVISION OF THESES AND RESEARCH FELLOWSHIPS

Current Ph.D. Theses: 2 *(supervisor C. Bellavita, FCAGLP and RUG, NL, co-supervisor L. West-Ocampo, FCAGLP)*
Ph.D. Thesis supervision: 1 *(E. Zubieta, FCAGLP, 10)*
Ph.D. These co-supervisions: 2 *(2023, co-promotor V. Peirano, RUG, Países Bajos; 2023, F. Fogantini, FCAGLP, 10)*
M.Sc. Theses: 3 *(sup. C. Bellavita, FCAGLP, 10; cosup. E. Zubieta, FCEN-UBA, 10; cosup. F. Fogantini, FCAGLP, 10)*
Undergrad Fellowships: 2 *(sup. C. Bellavita, CIN; co-sup. E. Zubieta, AAA)*
Bachelor Theses: 2 *(cosup. J. Roomer, U. Fontys and G. Boesjes, U. Groningen, the Netherlands)*

MEMBER OF THE JURY OF MASTER AND PHD THESES

Ph.D. Theses: 6 *(2019 in France, 2021 and 2023ab in the Netherlands, 2024 at FAMAF and 2026 at FCAGLP, Argentina)*
M.Sc. Theses: 4 *(1 in 2021, 3 in 2023, Argentina)*

MEMBER OF SCIENTIFIC RESEARCH PROJECTS

MC Exchanges - EU - *Overcoming challenges in the evolution and nature of massive stars (OCEANS, Member, 2025-2028)*
PID2022 - SPAIN *– Studying the Outflows of High-Energy Sources (SOHES-UJA, Member, 2023-2026)*
PIBAA CONICET - ARG *– Spectral-timing studies of QPOs in X-ray binaries: Comptonization models (PI, 2022-2025)*
PIP CONICET - ARG *– Observational and theoretical studies of X-ray binary systems (Co-PI, 2021-2025)*
PICT ANPCyT - ARG *– Observational and theoretical studies of supernova remnants and compact objects (2018-2021)*
PIP CONICET - ARG *– Phenomenological studies of X-ray sources in the Universe (2015-2017)*

TIME ALLOCATED IN ASTRONOMICAL FACILITIES

X-ray observatories *(XMM-Newton, Chandra, NuSTAR, NICER, Swift)*
Optical/IR observatories *(Gemini South, HST, CASLEO)*

MEMBER OF GROUPS AND PROFESSIONAL ASSOCIATIONS

IAU — *International Astronomical Union (2019-)* **aaa** — *Argentine Astronomical Society (2012-)*
AJA — *Argentine Young Academy (2022-)* **PuMA Collaboration** — *Pulsar Monitoring in Argentina (2017-)*
GARX — *Group of X-ray Astrophysics Argentina (2022-)* **LIGO/Virgo Collaboration** — *APC Group France (2019-2024)*
GARRA — *Group of Relativistic Astrophysics and Radioastronomy Argentina (2010-)*
AAVSO — *American Association of Variable Star Observers (2008-2018)*

PARTICIPATION IN THE UNIVERSITY COUNCIL

Graduated Superior Councillor *– UNLP (2014-2017)* **Student Directive Councillor** *– FCAGLP (2010-2011)*
Student Academic Councillor*– FCAGLP (2008-2010)*

REFEREE IN INTERNATIONAL JOURNALS

Nature *(2023),* **ApJ** *(2021-2026),* **A&A** *(2024-2026),* **MNRAS** *(2020-2024),*
JHEA *(2024-2026),* **RAA** *(2022),* **Frontiers** *(2023),* **NewAstronomy** *(2026)*